

Be the Hash Function.

a manual computation · five rounds · pencil required · no un-blending

FIG. 01 – ENCODE every letter has a number

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

SPECIMEN

CAT

copy each number from fig. 01

FIG. 02 – PAD the message is stamped with its own length

99

3

C A T STOP LENGTH

FIG. 03 – MIX each state out walks down to the next state in

state ← (state × 3 + n) mod 26 initial state = 7

N°	STATE IN	× 3 =	+ N =	MOD 26 → STATE OUT
01	7			
02				
03				
04				
05				

FIG. 04 – DIGEST final state → letter (fig. 01) – number

CAT → -

the fingerprint: small enough to compare, impossible to reverse.

One Letter.

hash CAR · identical procedure · observe what refuses to stay similar

FIG. 05 — MIX, AGAIN

n sequence: 3·1·18·99·3

state ← (state × 3 + n) mod 26

initial state = 7

N°	STATE IN	× 3 =	+ N =	MOD 26 → STATE OUT
01	7			
02				
03				
04				
05				

FIG. 06 — COMPARISON

copy CAT's digest from sheet N° 01

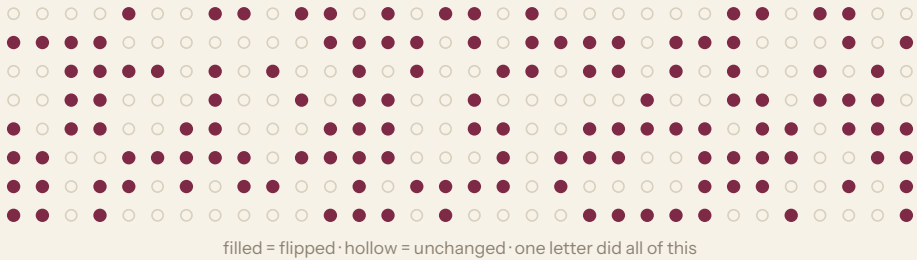
CAT → -

≠

CAR → -

FIG. 07 — THE AVALANCHE, OBSERVED

sha-256(CAT) xor sha-256(CAR) · 133 of 256 bits flipped · computed, not imagined



filled = flipped · hollow = unchanged · one letter did all of this

FIG. 08 — TWO MOVES FROM THE REAL MACHINE

sha-256 stirs with rotate, xor, add — you already added

XOR · THE DIFFERENCE DETECTOR

1 0 1 1
0 1 1 0

same: 0 · different: 1

ROTATE RIGHT 3 · THE BIT CAROUSEL

1 0 1 1 0 0 1 0

the last three bits wrap around to the front

One bit is enough.

The Hub.

one row per business key · identity, and nothing else

FIG. 09 – THE SOURCE LEDGER, OBSERVED six arrivals · strike every repeat with your pencil

01	CLM-101	arrived 2026-06-01
02	CLM-104	arrived 2026-06-01
03	CLM-101	arrived 2026-06-02
04	CLM-109	arrived 2026-06-03
05	CLM-104	arrived 2026-06-04
06	CLM-101	arrived 2026-06-05

FIG. 10 – THE HUB, RECORDED each distinct key enters exactly once · first arrival is remembered

BUSINESS KEY (PENCIL)	FIRST SEEN
	2026-06-01
	2026-06-01
	2026-06-03

FIG. 11 – A FOURTH ARRIVAL CLM-101 appears again on 06-09. how many hub rows now?



the hub does not count arrivals. it recognizes them.

A thing is itself only once.

The Link.

a relationship, recorded as a fact · nothing more is carried

FIG. 12 – THE REMITTANCE, OBSERVED which payment settled which claim

01	PMT-9001	settles	CLM-101	file 2026-06-12
02	PMT-9001	settles	CLM-104	file 2026-06-12
03	PMT-9002	settles	CLM-109	file 2026-06-14
04	PMT-9003	settles	CLM-101	file 2026-06-20

FIG. 13 – THE LINK, RECORDED one row per relationship · copy the pairs

CLAIM KEY (PENCIL)	PAYMENT KEY (PENCIL)

FIG. 14 – INTERROGATION PMT-9001 settled two claims · CLM-101 was settled twice

Does the link row store the payment amount?

☐ yes ☐ no – a satellite’s job

the link knows THAT they met. what happened between them is history — and history lives elsewhere.

Relationships are facts too.

The Satellite.

the diary of a business key · append, never overwrite

FIG. 15 – OBSERVATIONS OF CLM-101 three sightings, in the wild

01	status BILLED · amount 120	observed 2026-06-01
02	status DENIED · amount 120	observed 2026-06-09
03	status APPEALED · amount 120	observed 2026-06-21

FIG. 16 – THE SATELLITE, RECORDED load timestamps are given · enter each status in its row

LDTs	STATUS (PENCIL)
2026-06-01	
2026-06-09	
2026-06-21	

FIG. 17 – A QUESTION OF TIME the auditors ask about the past

As of 2026-06-15, the status of CLM-101 was:

take the newest row at or before the date — the pile, the timeline, the top card. you know this move.

History is kept, never overwritten.

The Hashdiff.

change detection by fingerprint · the blender returns

FIG. 18 – THREE DAYS, ONE CLAIM each day the source sends the status again

01	day 1 · status OK	hash it → fig. 19
02	day 2 · status OK	decide → fig. 20
03	day 3 · status NO	hash it, decide → fig. 21

FIG. 19 – FINGERPRINT DAY 1 mini blender · start at 7 · stop 99 · length 2

OK → n: 15 · 11 · 99 · 2				NO → n: 14 · 15 · 99 · 2 (for day 3)			
N°	IN	x3+n	MOD 26	N°	IN	x3+n	MOD 26
01	7			01	7		
02				02			
03				03			
04				04			

digest = final state, as letter-number

FIG. 20 – DAY 2 DECISION the source sends OK again

New fingerprint vs stored fingerprint — same or different? Then:

☐ insert new row

☐ skip — nothing changed

FIG. 21 – DAY 3 DECISION the source sends NO

Compare fingerprints once more. Then:

☐ insert new row

☐ skip — nothing changed

one comparison · any number of columns · this is the entire trick, and it is enough

Compare fingerprints, not paragraphs.

KEY · I : C3 A1 T20, ROUNDS 24 21 5 10 7, CAT+G-7, II : II: ROUNDS 24 21 3 4 15, CAR+O-15, XOR 1101, ROT 01010110
III: 3 ROWS (STILL 3) · IV: NO · V: DENIED · VI: OK+R-18 (10 15 14 18), NO+A-1 (9 16 17 1), DAY 2 SKIP, DAY 3 INSERT
VI · VI